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Experiential Learning Report

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**Topic: Smart UV Sterilization Safety System Using Multi-Layer
Protection Mechanisms**

BIO SAFETY STANDARDS AND ETHICS (BT242TC)

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ABSTRACT

Ultraviolet-C (UV-C) radiation is widely used for sterilization because of its ability to inactivate bacteria, viruses, fungi, and other harmful microorganisms without the use of chemical disinfectants. Despite its effectiveness, direct exposure to UV-C radiation can cause serious damage to human skin and eyes, making operational safety a major challenge in the deployment of UV-based sterilization systems. Most low-cost sterilization units primarily focus on disinfection while offering only limited protection against accidental human exposure.

This work presents a **Smart UV Sterilization Safety System** based on a multi-layer embedded safety framework that combines multiple independent protection mechanisms within a single control architecture. The proposed system is built around an Arduino Uno microcontroller and integrates a Passive Infrared (PIR) sensor for human presence detection, a magnetic reed switch for chamber door verification, a servo motor for automatic door locking, and a relay module for controlled operation of the UV-C lamp. Before initiating sterilization, the system verifies that the chamber door is securely closed and that no human presence is detected. It then activates visual and audible warning alerts, automatically locks the chamber door, and begins the sterilization cycle. During operation, the controller continuously monitors all safety conditions. If motion is detected or the chamber door is opened, the UV-C source is immediately switched off and the sterilization cycle is terminated through an emergency shutdown mechanism.

A prototype of the proposed system was developed and experimentally evaluated under different operating conditions. The results demonstrate reliable human detection, accurate door status verification, automatic control of the UV-C lamp, and immediate response to unsafe events. By integrating continuous monitoring, automated safety control, and real-time emergency protection into a low-cost embedded platform, the proposed system provides a practical and scalable solution for safe UV-C sterilization in hospitals, laboratories, educational institutions, offices, and household environments.

TABLE OF CONTENTS

1. Introduction
2. Problem Definition
 - Problem Statement
 - Background Information
3. Objectives
 - Primary Objectives
 - Secondary Objectives
4. Proposed Methodology
 - Research Gap Analysis
 - Workflow for proposed system
 - System Architecture
5. Project Execution
 - Planning and Design
6. Tools and Techniques Used
 - Hardware Components
 - Techniques Used
7. Results and Discussion
 - Expected Outcomes
 - Discussion
8. Proposed Prototype
 - Proposed Prototype Description
 - Development Process
 - Proposed Validation Strategy
 - Prototype Design
9. Conclusion
 - Summary
 - Personal Reflection
10. References

1. Introduction

Maintaining a clean and contamination-free environment is essential in healthcare facilities, laboratories, pharmaceutical industries, food processing units, educational institutions, and public spaces. The presence of harmful microorganisms on frequently used surfaces and equipment increases the risk of infection and disease transmission. Conventional sterilization methods, including chemical disinfectants, alcohol-based solutions, and thermal sterilization, are widely used to eliminate microbial contamination. Although these methods are effective in many situations, they have several practical limitations. Chemical disinfectants may leave residues on sensitive equipment, repeated manual cleaning is labour-intensive and time-consuming, and heat-based sterilization cannot be applied to all materials due to thermal sensitivity. These limitations have encouraged the adoption of alternative sterilization techniques that are both efficient and environmentally safe.

Ultraviolet-C (UV-C) radiation has become one of the most effective non-contact sterilization technologies because of its ability to inactivate bacteria, viruses, fungi, and other microorganisms without the use of chemicals. UV-C light, typically in the wavelength range of **200–280 nm**, damages the nucleic acids of microorganisms by disrupting their DNA and RNA structure. Once exposed to a sufficient UV-C dose, microorganisms lose their ability to reproduce, effectively preventing their growth and transmission. Since UV-C sterilization does not require chemical agents or produce secondary waste, it has been widely adopted in hospital sterilization chambers, laboratory equipment, water purification systems, air disinfection units, and medical device sterilization.

The importance of UV-C sterilization became particularly evident during the COVID-19 pandemic, when healthcare facilities and public institutions required rapid and contactless methods for disinfecting frequently used objects and enclosed spaces. This led to the development of portable sterilization boxes, autonomous UV disinfection robots, and enclosed UV-C chambers capable of reducing microbial contamination with minimal human intervention. While these systems successfully demonstrated the effectiveness of UV-C technology, many existing designs primarily concentrated on improving sterilization efficiency rather than ensuring operational safety.

Despite its advantages, UV-C radiation poses significant health risks when humans are directly exposed to it. Even short-term exposure can result in skin irritation, erythema, eye inflammation, and photokeratitis, while prolonged exposure may lead to more severe biological

damage. Consequently, every UV-C sterilization system must incorporate reliable safety mechanisms that prevent accidental exposure during operation. Existing low-cost sterilization systems often depend on manual switching, fixed timers, or a single safety sensor to control UV-C emission. These approaches may fail if the chamber door is opened during sterilization, if a person unexpectedly enters the operating area, or if the user unintentionally activates the UV source under unsafe conditions. Such limitations reduce the overall reliability of the sterilization process and increase the possibility of human exposure.

To address these challenges, this project proposes a **Smart UV Sterilization Safety System** that combines sterilization with multiple independent safety mechanisms within a single embedded platform. The system is designed around an Arduino Uno microcontroller and integrates a Passive Infrared (PIR) sensor for human presence detection, a magnetic reed switch for chamber door verification, a servo motor for automatic door locking, a relay module for UV-C lamp control, and visual and audible warning indicators. Before initiating sterilization, the controller verifies that the chamber door is securely closed and confirms that no human presence is detected. The system then activates warning alerts, automatically locks the chamber door, and begins the sterilization cycle. During operation, sensor inputs are continuously monitored, and any unsafe condition immediately triggers an emergency shutdown by switching off the UV-C lamp and terminating the sterilization process.

The proposed system demonstrates how embedded sensing, real-time monitoring, and automated control can be integrated to improve the operational safety of UV-C sterilization systems. Unlike many existing prototypes that rely on a single protection mechanism, the proposed design incorporates multiple safety layers that operate continuously throughout the sterilization cycle. This makes the system suitable for applications involving the sterilization of medical instruments, laboratory equipment, electronic devices, personal belongings, and other small objects while significantly reducing the possibility of accidental UV-C exposure.

2. Problem Definition

2.1 Problem Statement

The increasing use of **UV-C sterilization** in hospitals, laboratories, educational institutions, and public environments has highlighted the need for systems that provide both **effective disinfection** and **operational safety**. Although UV-C radiation is highly effective in eliminating bacteria, viruses, fungi, and other microorganisms, **direct exposure** to UV-C light can cause serious damage to **human skin** and **eye tissues**. Therefore, ensuring safe operation is one of the most important requirements in any UV-C sterilization system.

Most existing **low-cost UV sterilization systems** primarily focus on achieving efficient sterilization while providing only **limited safety mechanisms**. Many designs rely on **manual operation**, **timer-based control**, or a **single safety sensor**, making them vulnerable to unsafe situations. If the sterilization chamber is opened during operation or a person enters the sterilization area unexpectedly, harmful UV-C radiation may still be emitted, increasing the risk of accidental exposure. In addition, many commercially available systems do not provide **automatic door locking**, **continuous monitoring**, **real-time warning alerts**, or an **emergency shutdown mechanism** to respond immediately to unsafe operating conditions.

Another challenge is that advanced UV sterilization systems equipped with multiple safety features are often built using **industrial controllers**, **robotic platforms**, or expensive sensing technologies. These systems provide enhanced functionality but significantly increase the **overall cost**, making them unsuitable for deployment in **small laboratories**, **educational institutions**, **clinics**, and **household sterilization units**.

The problem addressed in this project is the development of a **low-cost embedded UV Sterilization Safety System** capable of combining **multiple independent safety mechanisms** into a single integrated platform. The system should continuously verify the **door status**, detect **human presence**, generate **visual and audible warning alerts**, perform **automatic door locking**, monitor the sterilization process in real time, and immediately execute an **emergency shutdown** whenever unsafe conditions are detected. By integrating these safety features within an **Arduino-based embedded system**, the proposed solution aims to improve user safety, increase system reliability, and provide a practical sterilization platform suitable for real-world applications.

2.2 Background Information (Literature Review)

2.2.1 UV-C Sterilization Technologies

Ultraviolet-C (UV-C) radiation has become one of the most widely adopted non-contact sterilization methods because of its ability to rapidly inactivate bacteria, viruses, fungi, and other harmful microorganisms without the use of chemical disinfectants. Research has demonstrated that UV-C radiation effectively damages the DNA and RNA of microorganisms, preventing replication and reducing the risk of disease transmission. Consequently, UV-C sterilization has been widely adopted in hospitals, laboratories, pharmaceutical industries, water purification systems, and enclosed sterilization chambers.

Several researchers have focused on developing compact and economical UV sterilization devices for practical deployment. Chong *et al.* proposed a programmable and low-cost UV room disinfection device that demonstrated reliable sterilization while significantly reducing implementation cost [1]. Similarly, Islam *et al.* investigated the combined use of heat and UV irradiation in a sterilization chamber and demonstrated improved disinfection performance for personal protective equipment and frequently used objects [8]. More recently, portable sterilization systems have been developed by combining multiple sterilization methods such as UV radiation, ozone generation, and ultrasonic cleaning to improve microbial inactivation under different operating conditions [24]. These studies establish the effectiveness of UV-C technology but primarily concentrate on sterilization efficiency rather than operational safety.

2.2.2 Robotic UV Disinfection Systems

To reduce manual intervention and improve coverage, researchers have increasingly integrated UV-C sterilization with autonomous robotic platforms. These systems enable automatic disinfection of hospital wards, laboratories, airports, and other public environments without requiring continuous human operation.

Several robotic systems have been developed with autonomous navigation and UV lamp control capabilities. The UV Sterilization Robot proposed by Nor *et al.* demonstrated automated movement for room sterilization using UV-C lamps mounted on a mobile platform [2]. Khatib *et al.* introduced UltraBot, an autonomous mobile robot capable of optimizing disinfection coverage through intelligent path planning [5]. Similarly, RoboCoV Cleaner incorporated autonomous navigation together with dual safety mechanisms to improve disinfection reliability in indoor environments [14]. Recent review articles have further

highlighted the growing importance of robotic UV sterilization systems and summarized various navigation, coverage planning, and control strategies developed over the past decade [15][26]. Although robotic systems significantly improve automation, many of them continue to rely on limited safety mechanisms and require expensive hardware for deployment.

2.2.3 Human Safety and Exposure Prevention

Since UV-C radiation is hazardous to human skin and eyes, numerous studies have investigated methods for preventing accidental exposure during sterilization. Human safety has become one of the most important research directions in the development of UV sterilization systems.

Researchers have proposed occupancy detection systems that automatically disable UV radiation whenever human presence is detected. Adeyemi *et al.* developed an automated workspace sterilization system that combines UV-C irradiation with occupancy detection to ensure that sterilization occurs only in unoccupied environments [16]. Similarly, Kumar *et al.* integrated motion sensing into a UV disinfection robot to automatically switch off the UV source whenever human movement was detected [3]. Recent studies have also explored the use of Far-UV-C radiation operating near 222 nm, which offers effective microbial inactivation while significantly reducing biological penetration into human skin and eye tissues [20]. These studies demonstrate important advances in UV safety; however, most systems implement only a single protective mechanism and do not integrate multiple safety layers within a unified embedded framework.

2.2.4 UV Monitoring and Intelligent Sterilization Systems

Recent research has shifted from simple UV lamp control towards intelligent sterilization systems capable of continuously monitoring environmental conditions and adapting their operation accordingly. Real-time monitoring improves sterilization reliability while reducing unnecessary UV exposure.

Tighe *et al.* developed a wireless ultraviolet sensing network capable of measuring UV dose during room disinfection, allowing operators to verify sterilization effectiveness while improving safety [13]. Kim *et al.* proposed an AI-based intelligent space sterilization system that monitors environmental conditions, analyses occupancy patterns, and dynamically controls air purification and surface sterilization modules [22]. Zhang *et al.* introduced an online microbial monitoring system integrated with UVA sterilization for hydroponic nutrient solutions, demonstrating simultaneous disinfection and real-time contamination monitoring

[23]. These intelligent systems improve operational efficiency but often require sophisticated sensing infrastructure, cloud connectivity, or specialized hardware that increases system complexity and cost.

Research Gap Identified

The literature survey indicates that significant progress has been made in the development of **UV-C sterilization technologies, autonomous disinfection robots, occupancy detection systems, and UV dose monitoring techniques**. Existing research has successfully demonstrated the effectiveness of UV-C radiation in eliminating microorganisms and reducing the spread of infectious diseases across healthcare, laboratory, and public environments. Several studies have focused on improving sterilization efficiency through robotic navigation, intelligent path planning, and automated control, while others have explored advanced technologies such as wireless UV sensing networks, AI-based environmental monitoring, and Far-UV-C radiation to enhance safety and performance.

Despite these advancements, a common limitation observed across the literature is that **sterilization efficiency is often given greater importance than operational safety**. Most low-cost UV sterilization systems employ only a **single safety mechanism**, such as a PIR sensor, occupancy detection, or timer-based control, to prevent accidental exposure. While these features provide a basic level of protection, they are not sufficient to address all possible unsafe operating conditions. For example, many systems do not verify whether the sterilization chamber is securely closed before UV activation, do not prevent users from opening the chamber during operation, and lack continuous monitoring throughout the sterilization cycle. As a result, accidental exposure to UV-C radiation remains a potential safety concern.

Another observation from the literature is that advanced safety features are generally found in **robotic or industrial-scale sterilization systems**, which depend on expensive sensors, autonomous mobile platforms, industrial controllers, or AI-based processing units. Although these systems offer improved functionality, their complexity and implementation cost make them unsuitable for educational institutions, small laboratories, clinics, and household sterilization applications where affordability and ease of maintenance are equally important.

The literature also reveals that **multiple safety mechanisms are rarely integrated into a single embedded system**. Features such as **human presence detection, door status verification, automatic door locking, warning alerts, continuous real-time monitoring, and emergency shutdown** are typically implemented independently in different systems rather

than working together as part of a coordinated safety framework. The absence of such integration reduces the overall reliability of the sterilization process and increases dependence on user awareness and manual supervision.

To address these limitations, the proposed **Smart UV Sterilization Safety System** introduces a **multi-layer embedded safety framework** that combines several independent protection mechanisms into a single low-cost architecture. The system continuously verifies the operating conditions before and during sterilization by integrating a **PIR sensor** for human presence detection, a **magnetic reed switch** for chamber door verification, a **servo motor** for automatic door locking, **visual and audible warning indicators**, and an **emergency shutdown mechanism** controlled by an **Arduino Uno**. Unlike many existing systems that focus primarily on sterilization performance, the proposed solution places equal emphasis on **human safety**, **system reliability**, and **practical implementation**, making it suitable for hospitals, laboratories, educational institutions, offices, and household sterilization chambers.

3. Objectives

3.1 Primary Objective

To design and develop a **Smart UV Sterilization Safety System** that enables effective UV-C sterilization while ensuring safe operation through continuous monitoring, automated control, and multiple integrated safety mechanisms to prevent accidental human exposure.

3.2 Secondary Objectives

- To design a **low-cost embedded UV-C sterilization system** using an Arduino Uno microcontroller for safe and reliable operation.
- To implement **human presence detection** using a Passive Infrared (PIR) sensor to prevent accidental exposure to UV-C radiation during sterilization.
- To incorporate a **magnetic reed switch** for continuous monitoring of the sterilization chamber door and ensure that UV-C radiation is activated only when the chamber is completely closed.
- To develop an **automatic door locking mechanism** using a servo motor to restrict access to the sterilization chamber while the UV-C lamp is in operation.
- To integrate **relay-based control** for safe switching of the UV-C lamp based on sensor inputs and predefined safety conditions.

- To provide **visual and audible warning indicators** that notify users before sterilization begins and during emergency situations.
- To implement a **real-time emergency shutdown mechanism** capable of immediately switching OFF the UV-C lamp whenever unsafe conditions such as human detection or door opening are identified.
- To validate the overall system by evaluating its functionality, safety response, reliability, and effectiveness under different operating conditions.
- To develop a **multi-layer embedded safety framework** that combines multiple independent protection mechanisms into a single integrated system for improved operational safety and reliability.

4. Proposed Methodology

The proposed **Smart UV Sterilization Safety System** is developed using a **multi-layer embedded safety framework** that combines sensing, decision-making, and automated control to ensure both effective UV-C sterilization and operator safety. Unlike conventional UV sterilization systems that primarily focus on microbial disinfection, the proposed methodology continuously evaluates the operating environment before and during sterilization to prevent accidental exposure to UV-C radiation.

The system is built around an **Arduino Uno microcontroller**, which acts as the central processing unit responsible for acquiring sensor inputs, executing safety verification, and controlling all output devices. The methodology follows a sequence of safety checks before initiating sterilization and continues to monitor the environment throughout the entire sterilization cycle.

Initially, the object to be sterilized is placed inside the sterilization chamber and the chamber door is closed. A **magnetic reed switch** continuously monitors the door status and confirms whether the chamber is completely sealed. If the door remains open or is not properly closed, the controller prevents the UV-C lamp from being activated and alerts the user through warning indicators. This verification ensures that UV radiation is never emitted while the chamber is accessible.

Once the door status is verified, the system activates the **Passive Infrared (PIR) sensor** to detect human presence near the sterilization chamber. The controller continuously analyses the sensor output to determine whether any movement is detected within the operating region. If

human presence is identified, the sterilization cycle is cancelled and the system remains in a safe standby state until the area becomes clear. This prevents accidental exposure to UV-C radiation before sterilization begins.

After both safety conditions are successfully verified, the system activates **visual and audible warning indicators** to inform nearby users that the sterilization process is about to begin. Following the warning period, a **servo motor** automatically locks the chamber door to prevent accidental opening during operation. The controller then energizes a **relay module**, which supplies power to the UV-C lamp and initiates the sterilization process for the predefined exposure duration.

Unlike timer-based sterilization systems, the proposed framework performs **continuous safety monitoring** throughout the sterilization cycle. The controller repeatedly checks the outputs of both the PIR sensor and the magnetic reed switch. If human presence is detected or the chamber door is opened while sterilization is in progress, the controller immediately deactivates the relay, switches OFF the UV-C lamp, activates the warning indicators, and terminates the sterilization cycle through an **emergency shutdown mechanism**. This real-time response significantly reduces the possibility of accidental human exposure and improves overall operational safety.

If no unsafe condition is detected during the sterilization interval, the controller switches OFF the UV-C lamp after the completion of the preset exposure time. The servo motor then unlocks the chamber door, allowing the sterilized object to be safely removed. The system subsequently returns to the standby state and waits for the next sterilization cycle.

The proposed methodology therefore integrates **human presence detection, door status verification, automatic door locking, relay-controlled UV-C operation, warning alerts, continuous monitoring, and emergency shutdown** into a unified embedded control framework. By combining these independent safety mechanisms, the system achieves reliable sterilization while maintaining a high level of user safety, making it suitable for applications in hospitals, laboratories, educational institutions, offices, and household environments.

4.1 Research Gap Analysis

Feature	Existing Systems	Proposed System
UV-C Sterilization	✓	✓
Human Presence Detection	Partial	✓
Door Status Verification	Partial	✓
Automatic Door Locking	×	✓
Warning Alerts	Partial	✓
Real-Time Monitoring	Partial	✓
Emergency Shutdown	Partial	✓
Multi-Layer Safety Framework	×	✓
Low-Cost Embedded Design	Partial	✓
Suitable for Small-Scale Deployment	Partial	✓

4.2 Workflow of the Proposed System

The workflow of the proposed **Smart UV Sterilization Safety System** is designed to ensure that UV-C sterilization is carried out only after all safety conditions have been verified. The system follows a sequential control process in which each stage must be successfully completed before proceeding to the next. During sterilization, the controller continuously monitors the operating conditions and immediately responds to any unsafe event through an emergency shutdown mechanism. The complete workflow is illustrated in **Figure 5.1**.

The operational sequence of the system is described below:

1. Object Placement

The object requiring sterilization is placed inside the sterilization chamber, and the user closes the chamber door before initiating the sterilization cycle.

2. Door Status Verification

A **magnetic reed switch** continuously monitors the chamber door. The Arduino controller verifies whether the door is completely closed before allowing the sterilization process to begin.

- If the door is open, the system remains in the standby state.
- The **UV-C lamp remains OFF** until proper door closure is confirmed.

3. Human Presence Detection

After confirming the door status, the system activates the **Passive Infrared (PIR) sensor** to detect human presence near the sterilization chamber.

- If motion is detected, sterilization is cancelled.
- The controller continuously waits until the surrounding area becomes clear before proceeding.

4. Warning and Safety Notification

Once all safety conditions are satisfied, the system activates **visual (LED)** and **audible (buzzer)** warning indicators to inform nearby users that the sterilization process is about to begin.

5. Automatic Door Locking

Following the warning period, a **servo motor** automatically locks the chamber door, preventing accidental access while the UV-C lamp is operating.

6. UV-C Sterilization Process

The Arduino controller energizes the **relay module**, which supplies power to the **UV-C lamp**. The sterilization cycle then begins for the predefined exposure duration.

7. Continuous Safety Monitoring

During sterilization, the controller continuously monitors all safety sensors to ensure that safe operating conditions are maintained.

The following conditions are repeatedly verified:

- Chamber door remains closed.
- No human presence is detected.
- Sterilization timer is operating normally.

8. Emergency Shutdown

If any unsafe condition is detected during sterilization, the controller immediately initiates the emergency protection sequence.

The following actions are performed automatically:

- UV-C lamp is switched OFF.
- Relay module is deactivated.
- Warning indicators are activated.
- Sterilization process is terminated.
- The system returns to a safe standby state.

9. Completion of Sterilization

If no safety violations occur, the sterilization cycle continues until the preset exposure time is completed. The controller then:

- Switches OFF the UV-C lamp.
- Unlocks the chamber door using the servo motor.
- Indicates successful completion through status LEDs.
- Returns the system to standby mode for the next sterilization cycle.

The above workflow ensures that **sterilization is performed only under verified safe conditions** while continuously monitoring the operating environment throughout the process. The integration of **door verification, human presence detection, automatic locking, warning alerts, continuous monitoring, and emergency shutdown** provides multiple layers of protection, significantly improving the safety and reliability of the proposed UV-C sterilization system.

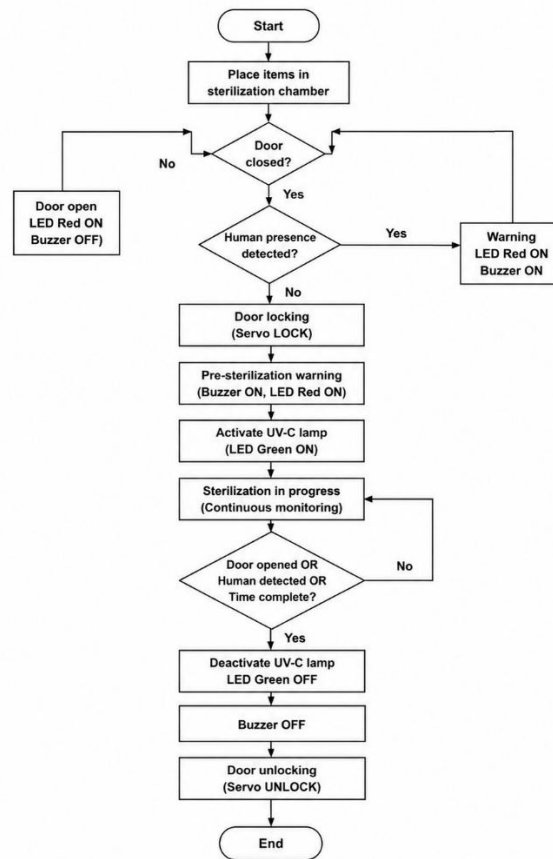


Figure 4.2.1: Workflow of the Proposed Smart UV Sterilization Safety System.

4.3 System Architecture

The proposed **Smart UV Sterilization Safety System** is designed using a modular embedded architecture that integrates sensing, control, decision-making, and safety mechanisms into a single platform. The architecture consists of four major functional layers: the **Input Layer**, **Processing Layer**, **Control Layer**, and **Output Layer**. These layers work together to ensure that sterilization is performed only under verified safe conditions while continuously monitoring the operating environment throughout the sterilization cycle.

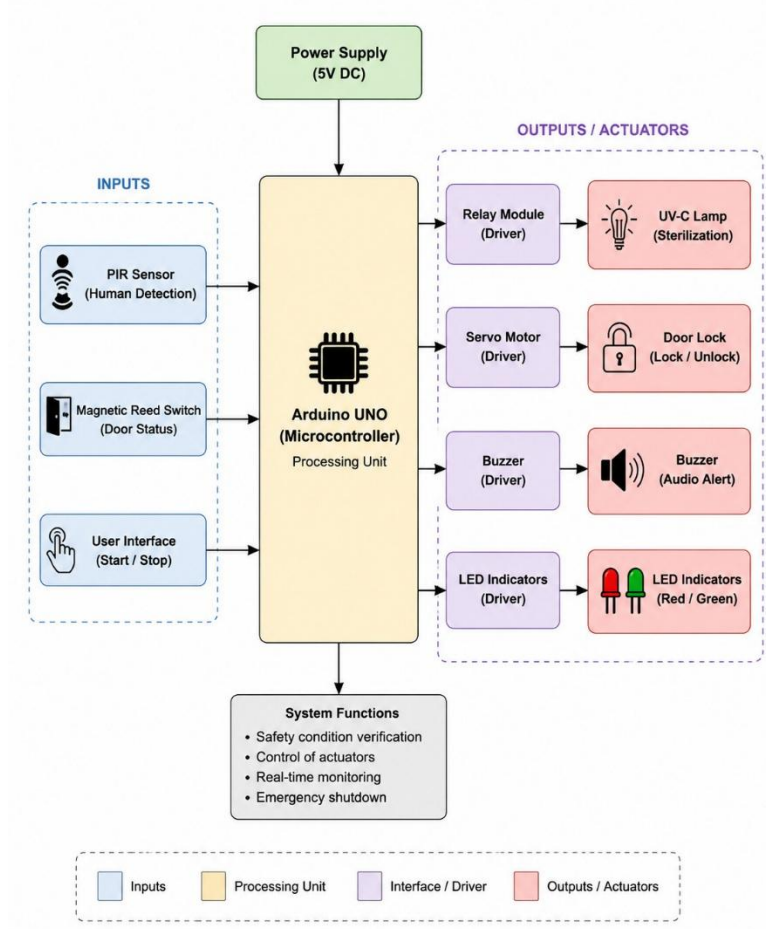


Figure 4.3.1: System Architecture

Input Layer

The Input Layer is responsible for acquiring real-time information about the operating environment and providing it to the controller for safety verification. It consists of the following sensing components:

• Passive Infrared (PIR) Sensor

The **PIR sensor** detects human movement around the sterilization chamber. Before sterilization begins, the sensor verifies that no person is present near the system. During sterilization, it continuously monitors the surroundings, and any detected movement immediately triggers the emergency shutdown mechanism.

• Magnetic Reed Switch

The **magnetic reed switch** continuously monitors the status of the sterilization chamber door. It confirms whether the chamber is completely closed before allowing the UV-C lamp to

operate. If the door is opened during sterilization, the controller immediately disables the UV-C lamp to prevent radiation leakage.

Processing Layer

The **Arduino Uno microcontroller** forms the central processing unit of the proposed system. It receives input signals from the PIR sensor and reed switch, processes the information according to the programmed safety algorithm, and generates appropriate control signals for all output devices.

The controller performs several important functions, including:

- Reading sensor inputs continuously.
- Verifying all safety conditions before sterilization.
- Controlling the sterilization timer.
- Operating the relay module.
- Controlling the servo motor.
- Activating warning indicators.
- Executing emergency shutdown whenever unsafe conditions are detected.

The Arduino therefore acts as the decision-making unit responsible for coordinating every stage of the sterilization process.

Control Layer

The Control Layer consists of the actuators responsible for implementing the controller's decisions.

• Relay Module

The relay module electrically isolates the low-voltage control circuit from the high-voltage UV-C lamp. Based on commands received from the Arduino, the relay switches the UV-C lamp ON or OFF while ensuring safe electrical operation.

• Servo Motor

The servo motor operates the automatic locking mechanism of the sterilization chamber. Once all safety conditions are verified, the controller commands the servo motor to lock the chamber

door before UV-C radiation is activated. After sterilization is completed or an emergency shutdown occurs, the servo unlocks the chamber, allowing safe access to the sterilized object.

Output Layer

The Output Layer provides sterilization functionality together with real-time user feedback regarding the operating status of the system.

• UV-C Lamp

The UV-C lamp serves as the primary sterilization unit. It emits ultraviolet radiation only after all safety conditions have been verified. The lamp remains active for the predefined sterilization duration unless an unsafe condition is detected.

• LED Indicators

Multiple LEDs are used to indicate different operating states of the system, including:

- System Ready
- Door Open
- Sterilization in Progress
- Sterilization Complete
- Emergency Shutdown

These indicators allow the user to quickly identify the current operating condition of the sterilization chamber.

• Buzzer

The buzzer provides audible alerts before sterilization begins and during emergency conditions. It warns nearby users that UV-C radiation is about to be activated or that an unsafe condition has occurred requiring immediate attention.

Overall System Operation

The system architecture enables all hardware components to operate as a coordinated embedded safety framework. Information acquired by the sensors is continuously processed by the Arduino Uno, which determines whether sterilization can safely proceed. Based on these

decisions, the controller operates the relay module, servo motor, LEDs, and buzzer in a predefined sequence.

Unlike conventional UV sterilization systems that rely on a single protective mechanism, the proposed architecture combines **human presence detection, door status verification, automatic door locking, warning alerts, continuous safety monitoring, and emergency shutdown** within a single embedded platform. This layered architecture significantly improves operational safety while maintaining reliable UV-C sterilization performance.

5. Project Execution

5.1 Planning and Design

The planning and design phase focused on developing a **Smart UV Sterilization Safety System** capable of performing effective UV-C sterilization while ensuring complete operational safety. Based on the literature survey, it was observed that many existing UV sterilization systems primarily emphasize sterilization efficiency and provide only limited protection against accidental human exposure. Therefore, the proposed system was planned with the objective of integrating multiple independent safety mechanisms into a single embedded platform.

The planning and design activities carried out during the project are described below.

Project Conceptualization

- Identified the need for a **safe and automated UV-C sterilization system** capable of minimizing accidental human exposure.
- Studied existing UV sterilization technologies and identified their limitations in terms of operational safety.
- Selected a **multi-layer embedded safety framework** as the foundation of the proposed system.

Requirement Analysis

The functional and safety requirements of the system were identified before hardware development.

Functional Requirements

- Perform effective UV-C sterilization inside an enclosed chamber.
- Automate the complete sterilization process.
- Monitor all safety conditions in real time.
- Provide user notifications throughout system operation.

Safety Requirements

- Verify chamber door closure before UV activation.
- Detect human presence near the sterilization chamber.
- Prevent accidental opening during sterilization.
- Generate warning alerts before UV activation.
- Execute immediate emergency shutdown whenever unsafe conditions are detected.

System Design

A modular embedded architecture was designed to divide the system into independent functional units.

The proposed architecture consists of:

- **Sensing Unit** for environmental monitoring.
- **Control Unit** for processing sensor information.
- **Safety and Monitoring Unit** for continuous verification.
- **Warning and Indication Unit** for user notifications.
- **UV Sterilization Unit** for disinfection.
- **Door Actuation Unit** for automatic chamber locking.

This modular approach simplified system integration while improving maintainability and reliability.

Hardware Planning

The hardware components were selected based on reliability, compatibility, availability, and implementation cost.

The planned hardware includes:

- **Arduino Uno** as the central controller.
- **Passive Infrared (PIR) Sensor** for human presence detection.
- **Magnetic Reed Switch** for chamber door monitoring.
- **Relay Module** for UV-C lamp control.
- **Servo Motor** for automatic door locking.
- **UV-C Lamp** as the sterilization source.
- **LED Indicators** for system status display.
- **Piezo Buzzer** for audible warning alerts.

Software Planning

The embedded software was planned to implement a sequential control algorithm capable of coordinating all hardware components.

The software was designed to perform the following operations:

- Acquire sensor inputs continuously.
- Verify all safety conditions before sterilization.
- Control the sterilization timer.
- Operate the relay module and servo motor.
- Generate visual and audible warnings.
- Perform continuous monitoring during sterilization.
- Execute emergency shutdown whenever required.

Workflow and System Architecture Design

Before implementation, detailed workflow and system architecture diagrams were prepared to define the interaction between all functional modules.

The workflow established the operational sequence of the system:

- Object placement.
- Door verification.
- Human presence detection.
- Warning generation.
- Automatic door locking.
- UV-C sterilization.
- Continuous monitoring.
- Emergency shutdown.
- Sterilization completion.

The system architecture was designed to clearly represent the communication between sensors, controller, actuators, and safety mechanisms.

Prototype Planning

A compact enclosed sterilization chamber was designed to demonstrate the proposed safety framework.

The prototype planning included:

- Proper positioning of the UV-C lamp.
- Sensor placement for reliable detection.
- Door locking arrangement using a servo motor.
- Routing of electrical connections.
- Safe installation of the control circuitry.
- Easy accessibility for testing and maintenance.

Testing Strategy

A comprehensive testing plan was prepared before system implementation to validate all functional and safety requirements.

The planned test cases included:

- Chamber door open before sterilization.
- Human presence detected before sterilization.
- Door opened during sterilization.
- Human movement detected during sterilization.
- Warning indicator operation.
- Automatic door locking.
- Timer-controlled sterilization.
- Emergency shutdown response.
- Complete sterilization cycle under normal conditions.

The planning and design phase established the overall framework for the proposed Smart UV Sterilization Safety System and ensured that every hardware and software component contributed to the primary objective of achieving **safe, reliable, and automated UV-C sterilization**. The systematic planning also simplified the implementation process and ensured consistency between the proposed methodology, system architecture, and final prototype.

Expected Advantages

The proposed framework is expected to provide the following advantages:

- Improved operator safety through multiple independent protection layers.
- Continuous monitoring of critical safety conditions.
- Reduced possibility of accidental UV-C exposure.
- Low-cost implementation using commercially available embedded hardware.
- Simple architecture suitable for educational institutions, laboratories, hospitals, and household sterilization chambers.
- Easy scalability for future IoT-based monitoring and intelligent control.

6. Tools and Techniques Used

6.1 Hardware Components

Arduino Uno

The **Arduino Uno** is proposed as the central controller of the system. It is responsible for acquiring sensor inputs, processing safety conditions, and controlling the operation of the UV-C lamp, servo motor, LEDs, and buzzer. The Arduino Uno is selected because of its simplicity, sufficient digital input/output pins, extensive library support, and ease of programming

Passive Infrared (PIR) Sensor

A **PIR sensor** is proposed for detecting human presence around the sterilization chamber. The sensor detects changes in infrared radiation emitted by moving objects and provides digital signals to the controller. Its primary purpose is to prevent accidental exposure to UV-C radiation by ensuring that sterilization begins only when no human movement is detected.

Magnetic Reed Switch

The **magnetic reed switch** is proposed for continuously monitoring the status of the sterilization chamber door. It provides a simple and reliable method of determining whether the chamber is securely closed before UV-C activation. The reed switch also assists in detecting unauthorized door opening during the sterilization process

Relay Module

A **relay module** is proposed to control the ON/OFF operation of the UV-C lamp. Since the UV-C lamp operates at a higher voltage than the Arduino controller, the relay provides electrical isolation while enabling safe switching of the sterilization unit.

Servo Motor

A **servo motor** is proposed to implement the automatic door locking mechanism. The servo receives position commands from the controller and locks the chamber door before sterilization begins. After the sterilization cycle is completed or interrupted, the servo unlocks the chamber to allow safe access.

UV-C Lamp

The **UV-C lamp** is the primary sterilization component of the proposed system. It emits ultraviolet radiation in the germicidal wavelength range, enabling effective inactivation of microorganisms. The lamp is designed to operate only after all safety conditions have been verified.

LED Indicators and Buzzer

LED indicators and a piezo buzzer are proposed to provide visual and audible feedback regarding the operating status of the system. These indicators inform users about standby mode, safety warnings, sterilization progress, completion of sterilization, and emergency shutdown events.

6.2 Techniques Used

Embedded System Design

The proposed framework follows an **embedded system design** approach in which sensing, processing, and control functions are integrated within a dedicated microcontroller. This approach enables real-time decision making while maintaining a compact and economical system architecture.

Multi-Layer Safety Framework

The proposed system employs a **multi-layer safety framework**, where multiple independent safety mechanisms operate together to minimize accidental UV-C exposure. Instead of relying on a single safety feature, the framework combines human presence detection, door verification, automatic locking, warning alerts, continuous monitoring, and emergency shutdown into a unified control strategy.

Sensor-Based Monitoring

The framework adopts **sensor-based monitoring** to continuously evaluate environmental conditions before and during sterilization. Real-time data from the PIR sensor and magnetic reed switch are processed by the controller to determine whether sterilization can safely continue.

Relay-Based Switching

The UV-C lamp is proposed to be controlled using **relay-based switching**, allowing safe electrical isolation between the low-voltage controller and the high-voltage sterilization unit.

This technique improves system safety and protects the embedded controller from electrical faults.

Real-Time Decision Making

The proposed framework utilizes **real-time decision making**, where the controller continuously processes sensor inputs and responds immediately to unsafe operating conditions. This enables rapid execution of emergency shutdown whenever human presence or door opening is detected.

Sequential Control Algorithm

A **sequential control algorithm** is proposed to coordinate the complete sterilization process. The algorithm executes safety verification, warning generation, automatic door locking, UV-C activation, continuous monitoring, and emergency shutdown in a predefined sequence to ensure reliable and safe system operation.

7. Results and Discussion

7.1 Expected Outcomes

The proposed **Smart UV Sterilization Safety System** is expected to provide a safe and reliable framework for UV-C sterilization by integrating multiple independent safety mechanisms into a single embedded architecture. Unlike conventional UV sterilization systems that primarily focus on microbial disinfection, the proposed framework emphasizes both **sterilization effectiveness** and **operator safety** through continuous monitoring and automated decision-making.

The expected outcomes of the proposed framework are summarized below.

Safe UV-C Sterilization

The proposed framework is expected to ensure that UV-C radiation is emitted only after verifying all predefined safety conditions. By combining chamber door verification and human presence detection, the system aims to eliminate accidental activation of the UV-C lamp under unsafe operating conditions.

Reliable Human Presence Detection

The integration of a **Passive Infrared (PIR) sensor** is expected to provide continuous monitoring of the area surrounding the sterilization chamber. Human presence detected before sterilization is expected to prevent system activation, while movement detected during operation is expected to immediately trigger the emergency shutdown mechanism.

Accurate Door Status Verification

The proposed **magnetic reed switch** is expected to continuously monitor the chamber door and verify that it remains securely closed throughout the sterilization process. This feature is intended to prevent UV-C radiation leakage caused by accidental or unauthorized opening of the chamber.

Automatic Door Locking

The proposed **servo motor-based locking mechanism** is expected to restrict access to the sterilization chamber during operation. Automatic locking is intended to reduce dependence on manual user intervention and improve overall operational safety.

Continuous Safety Monitoring

Unlike conventional timer-based sterilization systems, the proposed framework is expected to continuously evaluate safety conditions throughout the sterilization cycle. Real-time monitoring of sensor inputs is intended to enable immediate detection of unsafe operating conditions.

Emergency Shutdown Capability

The proposed emergency shutdown mechanism is expected to rapidly deactivate the UV-C lamp whenever unsafe conditions such as human presence or chamber door opening are detected. This feature is intended to significantly reduce the possibility of accidental UV-C exposure.

Improved Operational Safety

By integrating **human presence detection, door verification, automatic locking, warning alerts, continuous monitoring, and emergency shutdown**, the proposed framework is expected to provide multiple layers of protection that improve the overall safety and reliability of UV-C sterilization systems.

Low-Cost Embedded Solution

The proposed framework utilizes commercially available embedded hardware, making it suitable for economical implementation. The use of an Arduino Uno and low-cost sensing modules is expected to reduce implementation cost while maintaining reliable system performance.

Potential Application Areas

The proposed architecture is expected to be adaptable for use in:

- Hospitals and healthcare facilities
- Clinical laboratories
- Educational institutions
- Pharmaceutical industries
- Offices and commercial establishments
- Household sterilization chambers

The modular architecture also provides flexibility for future enhancements, including IoT-based remote monitoring, UV dose measurement, cloud-based data logging, and AI-assisted human detection.

7.2 Discussion

The literature survey indicates that most existing UV sterilization systems emphasize sterilization performance while implementing only limited safety mechanisms. Some systems provide occupancy detection, whereas others incorporate robotic navigation or UV dose monitoring. However, very few studies integrate **human presence detection, door verification, automatic door locking, continuous safety monitoring, warning alerts, and emergency shutdown** within a single low-cost embedded platform.

The proposed framework addresses this limitation by introducing a **multi-layer embedded safety architecture** in which multiple independent protection mechanisms operate together throughout the sterilization cycle. This layered approach is expected to improve operational safety without significantly increasing system complexity or cost.

Although the framework has not yet been implemented experimentally, the proposed architecture provides a strong foundation for future prototype development and practical

deployment. The modular design allows additional technologies such as IoT connectivity, UV intensity monitoring, AI-based occupancy detection, and predictive maintenance to be incorporated without major architectural modifications. Consequently, the proposed framework represents a scalable and practical approach toward the development of next-generation intelligent UV-C sterilization systems.

8. Proposed Prototype

8.1 Proposed Prototype Description

The proposed prototype is designed as a compact enclosed **UV-C sterilization chamber** integrated with multiple safety mechanisms to ensure effective sterilization while preventing accidental human exposure to ultraviolet radiation. The prototype follows a **multi-layer embedded safety framework**, where sensing, control, and safety modules operate together under the supervision of an Arduino Uno microcontroller.

The prototype is intended to consist of a UV-C lamp mounted inside an enclosed chamber, ensuring that ultraviolet radiation remains confined within the sterilization space. A **Passive Infrared (PIR) sensor** is proposed to monitor human presence around the chamber, while a **magnetic reed switch** continuously verifies the status of the chamber door. The controller receives these sensor inputs and determines whether the sterilization process can safely begin.

To enhance operational safety, the prototype incorporates a **servo motor** that automatically locks the chamber door before UV-C activation. A **relay module** is proposed to control the UV-C lamp, providing electrical isolation between the embedded controller and the high-voltage sterilization unit. Visual status LEDs and a piezo buzzer are also included to provide continuous feedback regarding system operation, warning notifications, sterilization progress, and emergency conditions.

The proposed prototype is intended to provide a practical demonstration of the integrated safety framework and can serve as a foundation for future prototype development and experimental validation.

8.2 Development Process

The development of the proposed Smart UV Sterilization Safety System is planned through a structured engineering approach to ensure systematic design and future implementation. The proposed development process consists of the following stages.

Requirement Analysis

- Study existing UV-C sterilization systems.
- Identify limitations related to operational safety.
- Define functional and safety requirements for the proposed framework.

System Design

- Design the overall system architecture.
- Develop the operational workflow.
- Design the embedded safety framework.
- Prepare the control algorithm for system operation.

Component Selection

- Select suitable embedded hardware components.
- Choose appropriate sensing devices for human detection and door monitoring.
- Select control and indication devices based on functionality and cost.

Software Development

- Develop the embedded control algorithm using the Arduino IDE.
- Integrate sensor monitoring, relay control, servo control, and warning mechanisms.
- Program the emergency shutdown logic.

System Integration

- Interface all hardware modules with the Arduino Uno.
- Establish communication between sensors, controller, and output devices.
- Verify proper interaction among all system components.

Testing and Validation

- Perform functional testing under different operating conditions.
- Verify the operation of all safety mechanisms.

- Evaluate system reliability and operational safety.
- Analyse the effectiveness of the proposed safety framework.

This structured development process provides a clear roadmap for converting the proposed framework into a fully functional prototype in future work.

8.3 Proposed Validation Strategy

The proposed framework is intended to be validated through a combination of **functional testing**, **safety verification**, and **performance evaluation**. The objective of the validation strategy is to confirm that every safety mechanism operates according to the proposed control algorithm while ensuring reliable UV-C sterilization.

The proposed validation includes the following test scenarios.

Test Scenario	Expected Behaviour
Chamber door open before sterilization	UV-C lamp remains OFF
Chamber door closed	System proceeds to safety verification
Human presence detected before sterilization	Sterilization is not initiated
No human detected	System proceeds to warning stage
Warning indication	LEDs and buzzer activate before UV-C operation
Automatic door locking	Chamber door locks before UV-C activation
UV-C sterilization	Lamp operates only after all safety conditions are satisfied
Door opened during sterilization	Emergency shutdown is activated immediately
Human detected during sterilization	UV-C lamp switches OFF instantly
Sterilization timer completed	UV-C lamp turns OFF and door unlocks

The proposed validation strategy is also intended to assess:

- Reliability of human presence detection.
- Accuracy of chamber door monitoring.
- Response time of the emergency shutdown mechanism.
- Correct operation of the relay-controlled UV-C lamp.
- Proper functioning of the automatic door locking mechanism.
- Overall reliability of the multi-layer embedded safety framework.

Successful validation of these parameters would demonstrate the feasibility and effectiveness of the proposed architecture for safe UV-C sterilization.

8.4 Prototype Design

The proposed prototype follows a **modular embedded architecture**, allowing each subsystem to perform a dedicated function while interacting with the central controller. The modular design improves scalability, maintainability, and future expandability.

The proposed design consists of four major subsystems:

Input Subsystem

The input subsystem acquires information from the operating environment.

It consists of:

- Passive Infrared (PIR) Sensor
- Magnetic Reed Switch

These sensors continuously provide information regarding human presence and chamber door status.

Control Subsystem

The control subsystem consists of the **Arduino Uno**, which functions as the central decision-making unit.

The controller is designed to:

- Read sensor inputs.
- Verify safety conditions.

- Execute the control algorithm.
- Coordinate all output devices.
- Monitor system operation continuously.

Safety and Actuation Subsystem

This subsystem is responsible for executing controller decisions.

It includes:

- Relay Module
- Servo Motor

The relay controls the UV-C lamp, while the servo motor performs automatic locking and unlocking of the chamber door.

Output Subsystem

The output subsystem provides sterilization and user notification.

It consists of:

- UV-C Lamp
- LED Indicators
- Piezo Buzzer

The UV-C lamp performs sterilization, while the LEDs and buzzer provide visual and audible information regarding system status.

The proposed prototype is designed to operate according to the multi-layer embedded safety framework presented in this report. By integrating **human presence detection, door verification, automatic door locking, warning alerts, continuous monitoring, and emergency shutdown** into a unified architecture, the proposed design aims to provide a practical, reliable, and economical solution for safe UV-C sterilization.

9. Conclusion

9.1 Summary

This project proposed a **Smart UV Sterilization Safety System** based on a **multi-layer embedded safety framework** for improving the safe operation of UV-C sterilization systems. Through an extensive literature survey, it was observed that many existing UV sterilization systems primarily focus on sterilization efficiency while providing only limited safety mechanisms to prevent accidental human exposure.

To address this limitation, the proposed framework integrates **human presence detection, door status verification, automatic door locking, warning alerts, continuous real-time monitoring**, and an **emergency shutdown mechanism** within a single embedded architecture. The proposed design utilizes an **Arduino Uno** as the central controller along with a **Passive Infrared (PIR) sensor, magnetic reed switch, relay module, servo motor, LED indicators**, and a **buzzer** to coordinate the safety and sterilization processes.

The proposed methodology, workflow, system architecture, and control algorithm provide a structured approach for developing a safe and economical UV-C sterilization system. The modular nature of the framework also allows future enhancements such as IoT-based monitoring, UV dose measurement, and intelligent control techniques to be incorporated without significant modifications to the overall architecture.

Overall, the proposed framework presents a practical foundation for the future development of intelligent UV-C sterilization systems where both **sterilization effectiveness** and **operator safety** are treated as equally important design objectives.

9.2 Personal Reflection

Working on this project helped us understand the importance of combining **engineering design** with **safety considerations** while proposing solutions for real-world problems. The literature review exposed us to different approaches used in UV-C sterilization, robotic disinfection, occupancy detection, UV monitoring, and embedded safety systems, enabling us to critically analyse existing research and identify the limitations that motivated this work.

The project also improved our understanding of **embedded system design, sensor integration**, and **system architecture development**. Preparing the workflow, system

architecture, control algorithm, and proposed framework strengthened our ability to design structured engineering solutions and represent them through technical documentation.

In addition, this project enhanced our skills in **technical writing**, **research methodology**, and **literature analysis**. Studying scientific publications and comparing different design approaches helped us understand how research gaps are identified and how proposed solutions are formulated based on existing work.

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